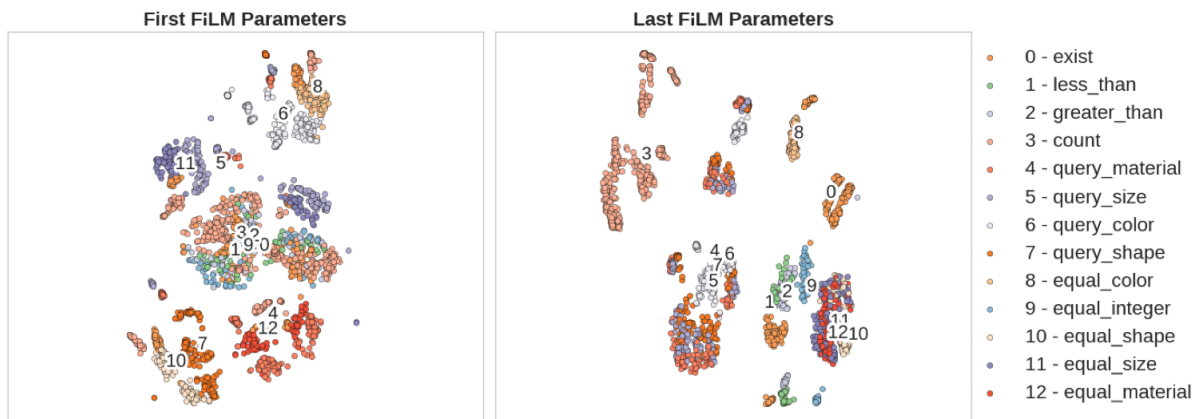
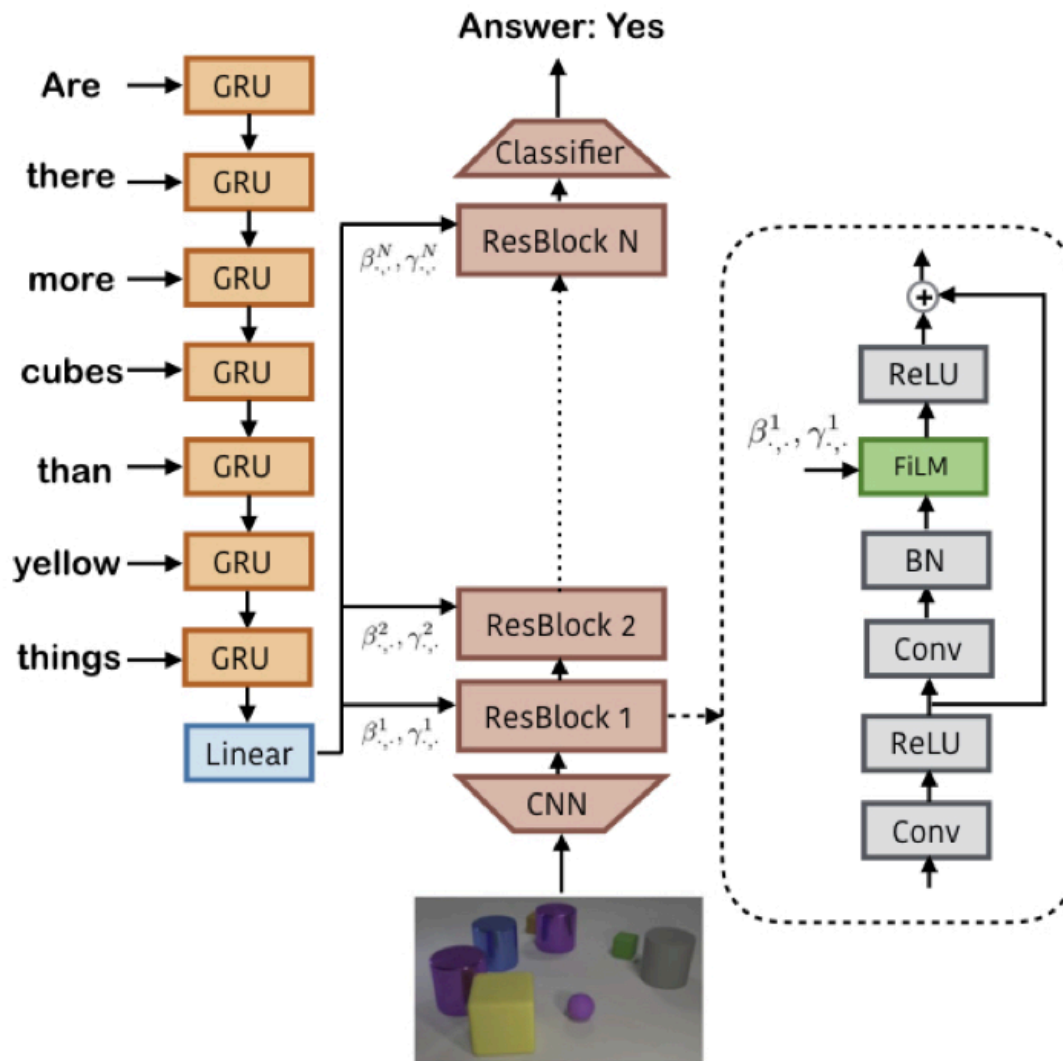


13th Meeting 03/25

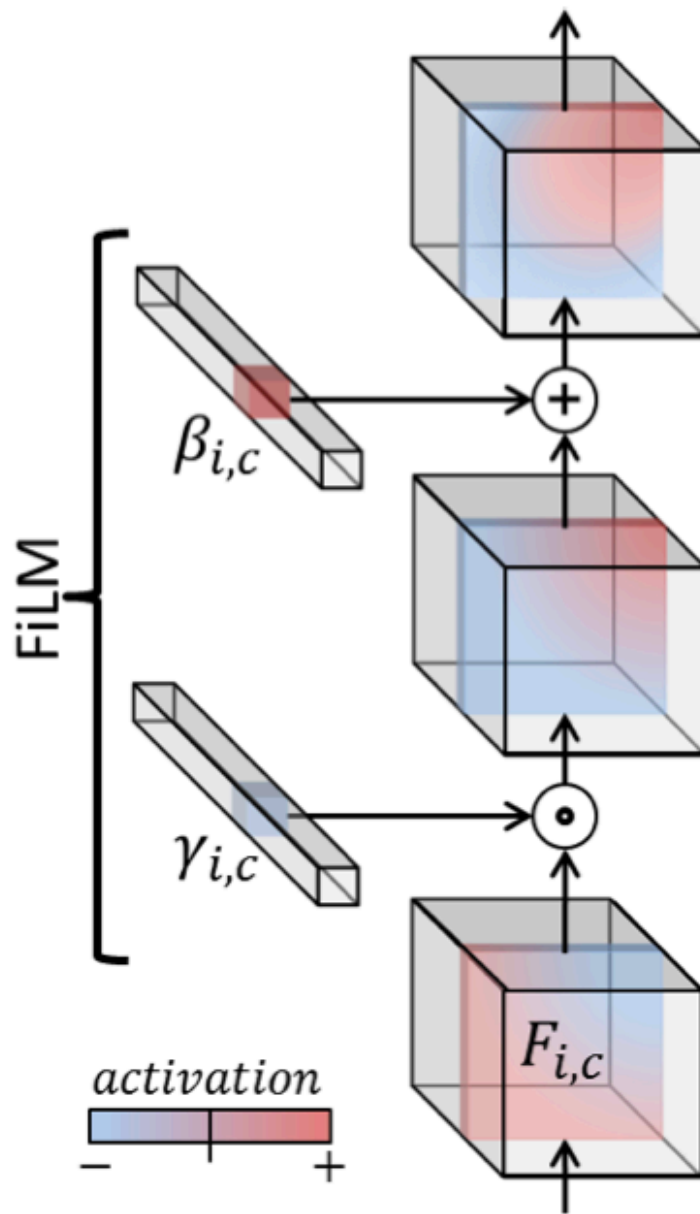
- The problem is, there is some activity that looks really similar and can't be differentiate only in the visual location. Need to given the clue of something.
- If just only by the visual, there will be a lot of ambiguity
- the class is so imbalance, just like real life.
- simultaneous hypothesis is to make the activity head consumes both input, forcing the network to learn
- proposed is hope to the model operates $\frac{1}{3}$ of its power, fundamentally richer, and can be kinematically learn

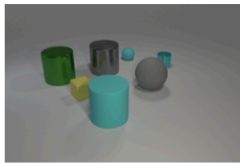


$$\gamma_{i,c} = f_c(\mathbf{x}_i) \quad \beta_{i,c} = h_c(\mathbf{x}_i), \quad (1)$$

where $\gamma_{i,c}$ and $\beta_{i,c}$ modulate a neural network's activations $\mathbf{F}_{i,c}$, whose subscripts refer to the i^{th} input's c^{th} feature or feature map, via a feature-wise affine transformation:

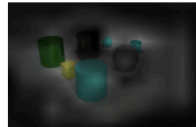
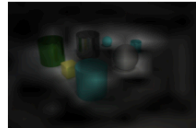
$$FiLM(\mathbf{F}_{i,c}|\gamma_{i,c}, \beta_{i,c}) = \gamma_{i,c}\mathbf{F}_{i,c} + \beta_{i,c}. \quad (2)$$



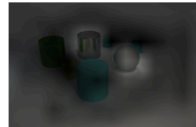
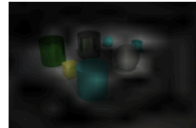


Feature 14 - Block 1

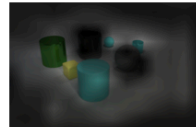
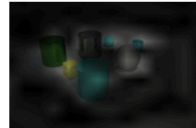
Before FiLM
After FiLM



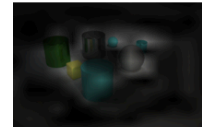
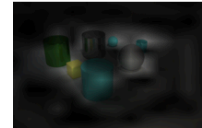
Q: What is the color of the large rubber cylinder? **A:** Cyan



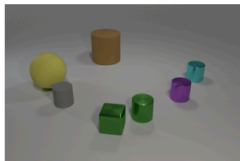
Q: What is the color of the large rubber sphere? **A:** Gray



Q: What is the color of the cube? **A:** Yellow

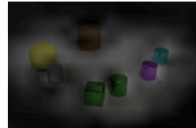


Q: How many cylinders are there? **A:** 4

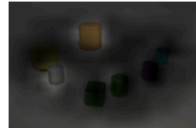


Feature 14 - Block 1

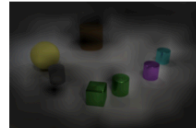
Before FiLM
After FiLM



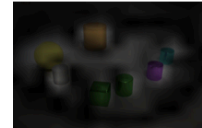
Q: What is the color of the large rubber cylinder? **A:** Yellow



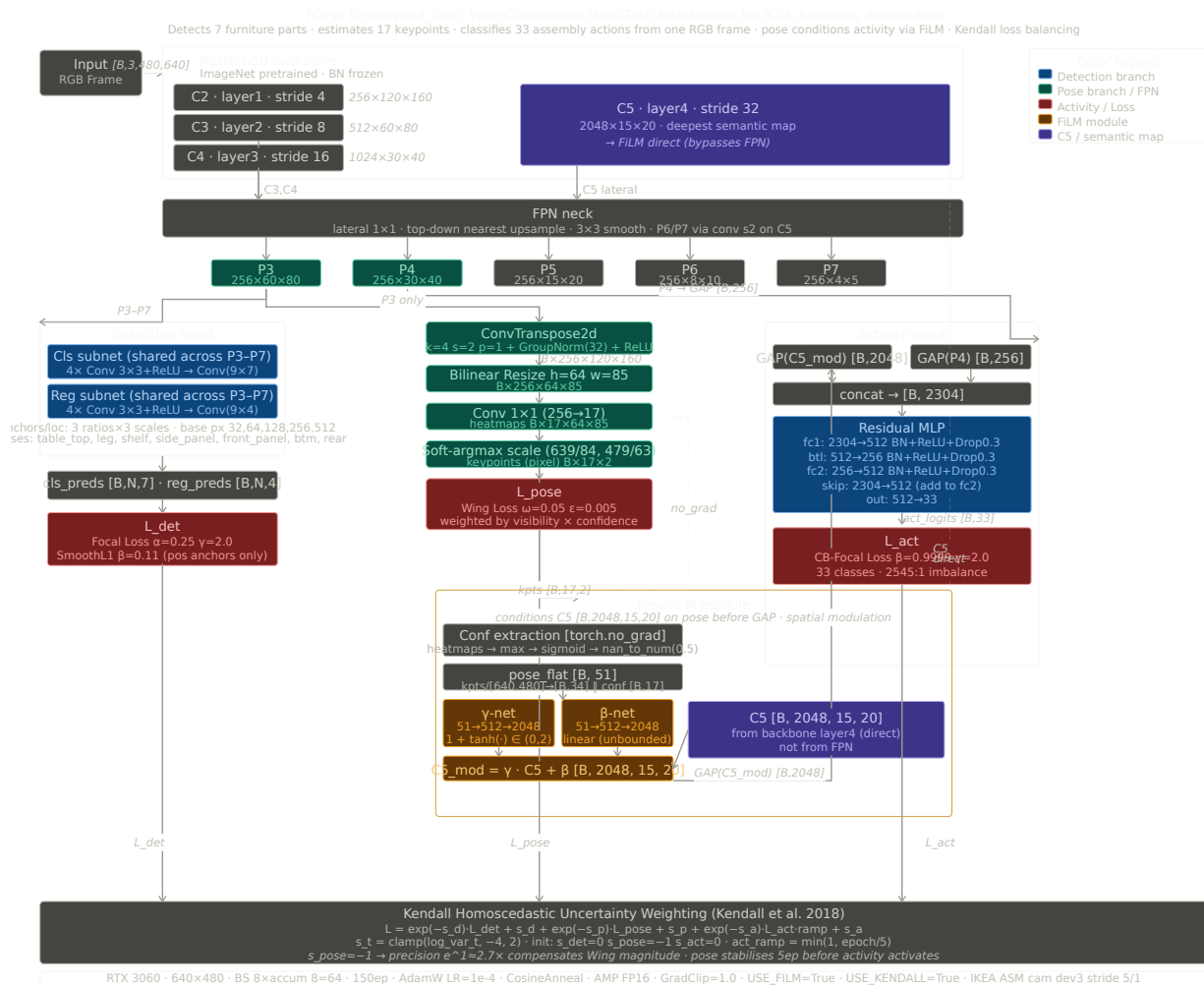
Q: What is the color of the large rubber sphere? **A:** Gray



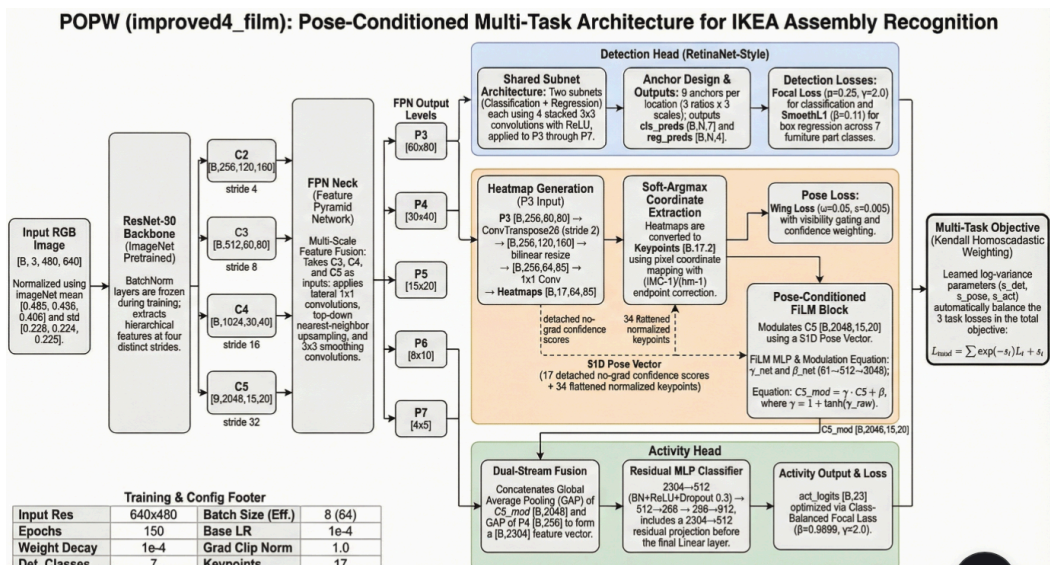
Q: What is the color of the cube? **A:** Yellow



Q: How many cylinders are there? **A:** 4



take the implementation attention in the activity head



Class-balanced weight (Cui et al., 2019)

Notation.

- c Class index $c \in \{0, \dots, C - 1\}$.
- n_c Number of training samples with label c (from dataset statistics passed to `set_class_counts`).
- β `CB_BETA` = 0.9999.
- E_c Effective number of samples: $E_c = (1 - \beta^{n_c}) / (1 - \beta)$ for $n_c > 0$; if $n_c = 0$, implementation uses `effective = 1` for that class in the NumPy branch.
- $\tilde{w}_c$ Raw inverse effective weight $\tilde{w}_c = 1 / \max(E_c, 10^{-8})$.
- w_c Normalized class weight stored in buffer: $\tilde{w}_c$ scaled so $\sum_{c'} w_{c'} = C$ (code: `weights / weights.sum() * num_classes`).

$$E_c = \frac{1 - \beta^{n_c}}{1 - \beta}, \quad \tilde{w}_c = \frac{1}{\max(E_c, \epsilon)}, \quad w_c = \frac{\tilde{w}_c}{\sum_{c'} \tilde{w}_{c'}} \cdot C$$

Here ϵ denotes the numerical floor `1e-8` in code. The tensor `class_weights[c]` equals w_c after `set_class_counts`.

Class-balanced weight (Cui et al., 2019)

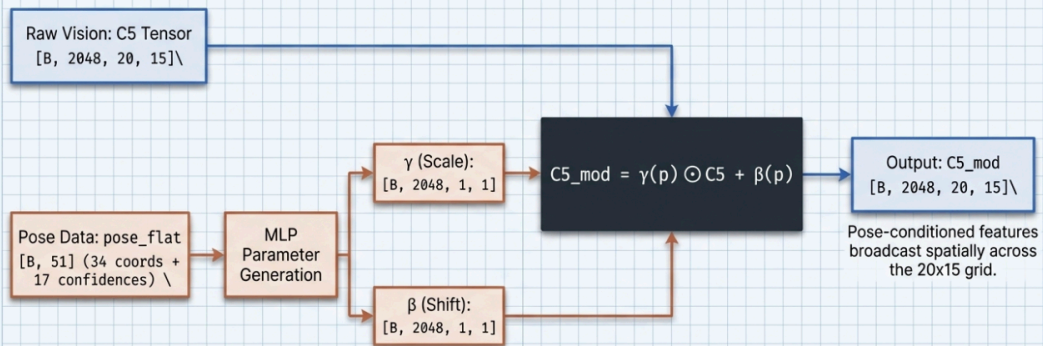
Notation.

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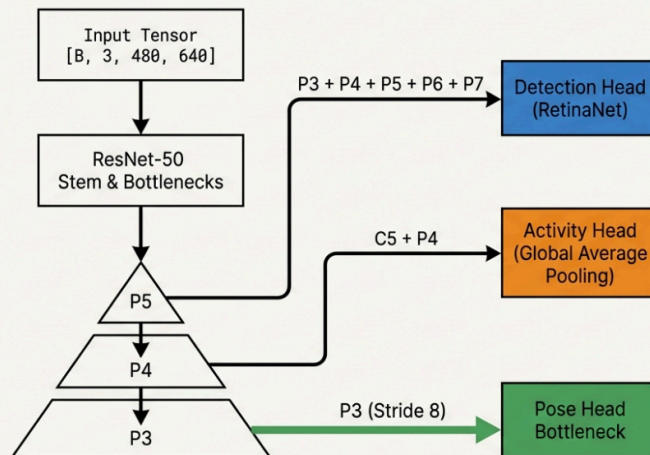
Here ϵ denotes the numerical floor `1e-8` in code. The tensor `class_weights[c]` equals w_c after `set_class_counts`.

Mathematical Zoom-In: Feature-wise Linear Modulation (FiLM) Mechanism



Critical Context: FiLM modulates raw C5 ONLY — immediately BEFORE the activity branch reads from it. **Detection and Pose heads remain entirely unaffected** by this modulation.

Routing Dense Pixels to Task-Specific Bottlenecks



Shared Representation Bottlenecks

- The architecture utilizes a ResNet-50 backbone with an FPN neck to serve multi-scale representations.
- RetinaNet anchors consume the full pyramid for scale-invariant object detection.
- The Activity head pools semantic-heavy C5 and P4 features for global scene context.

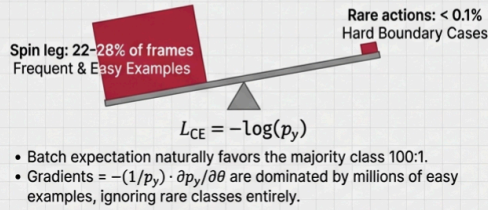
The P3 Pathway

- The Pose Head isolates the P3 feature map $[B, 256, 80, 60]$.
- Operating at 1/8 of the input spatial resolution.
- Provides the optimal balance of high-level body semantics and fine-grained spatial acuity required for accurate joint localization.

Class-Balanced Focal Loss: Handling Heavy-Tailed Action Distributions

Deconstructing the dual-pathology of the IKEA ASM dataset and the mathematical interventions required for stable convergence.

THE PATHOLOGY: STANDARD CE COLLAPSE



THE SYNTHESIS: UNIFIED CURE

$$L_{CB-Focal} = \underbrace{[-\alpha_y]}_{\text{Handles Volume}} \cdot \underbrace{[(1-p_y)^2]}_{\text{Handles Confidence}} \cdot \underbrace{[\log(p_y)]}_{\text{Base CE Loss}}$$

TRACK A: MACRO INTERVENTION (CLASS REWEIGHTING)

Concept: Effective Number of Samples (Cui et al.)

$$E_c = \frac{1 - \beta^{n_c}}{1 - \beta} \rightarrow \alpha_c = \frac{1 - \beta}{1 - \beta^{n_c}}$$

At $\beta = 0.9999$, $E_\infty = 10,000$. This perfectly saturates the overrepresented "NA" class ($\leq 10,000$ samples) without zeroing its weight.

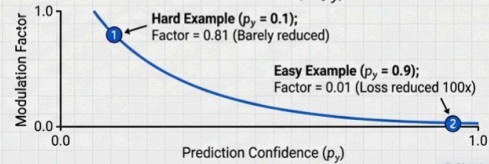
$n_c = 50$	$\rightarrow \alpha \approx 0.020$
$n_c = 500$	$\rightarrow \alpha \approx 0.002$
$n_c = 50000$	$\rightarrow \alpha \approx 0.0001$ (Saturated)

200:1 ratio forces network to respect rare classes

TRACK B: MICRO INTERVENTION (FOCAL MODULATION)

Concept: Easy-Example Dampener (Lin et al.)

Modulation factor $= (1-p_y)^{\gamma}$



The Theory: Heteroscedastic Uncertainty for Multi-Task Balancing.

Regression / Pose

$$p(y|f(x)) = \mathcal{N}(f(x), \sigma^2)$$

σ_t (Learned Observation Noise)

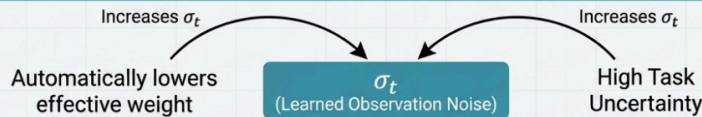
$$L = \frac{1}{2\sigma^2} \|y - f(x)\|^2 + \log \sigma$$

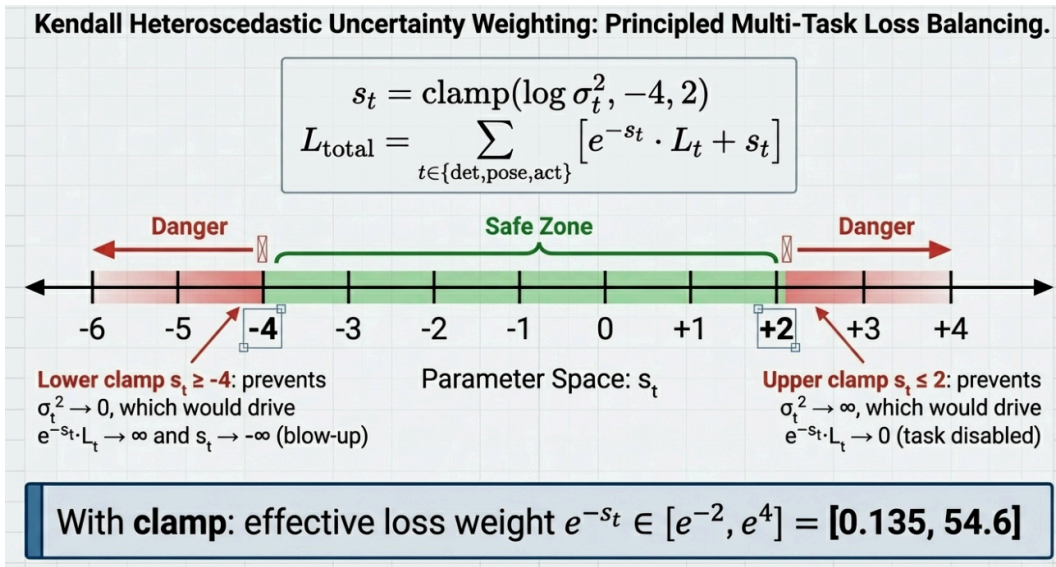
Classification / Act & Det

$$p(y|f(x)) = \text{Softmax}\left(\frac{f(x)}{\sigma^2}\right)$$

σ_t (Learned Observation Noise)

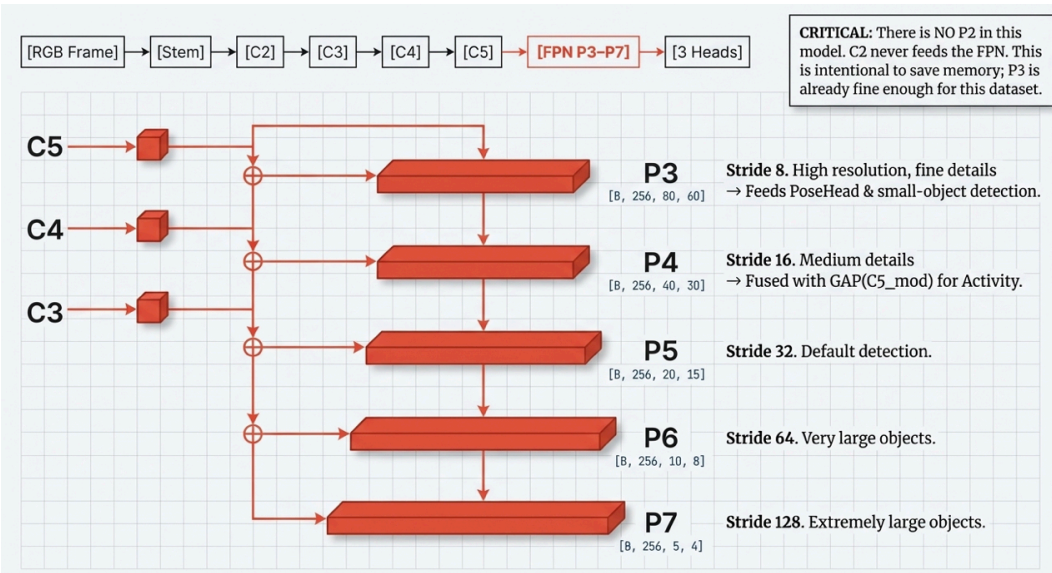
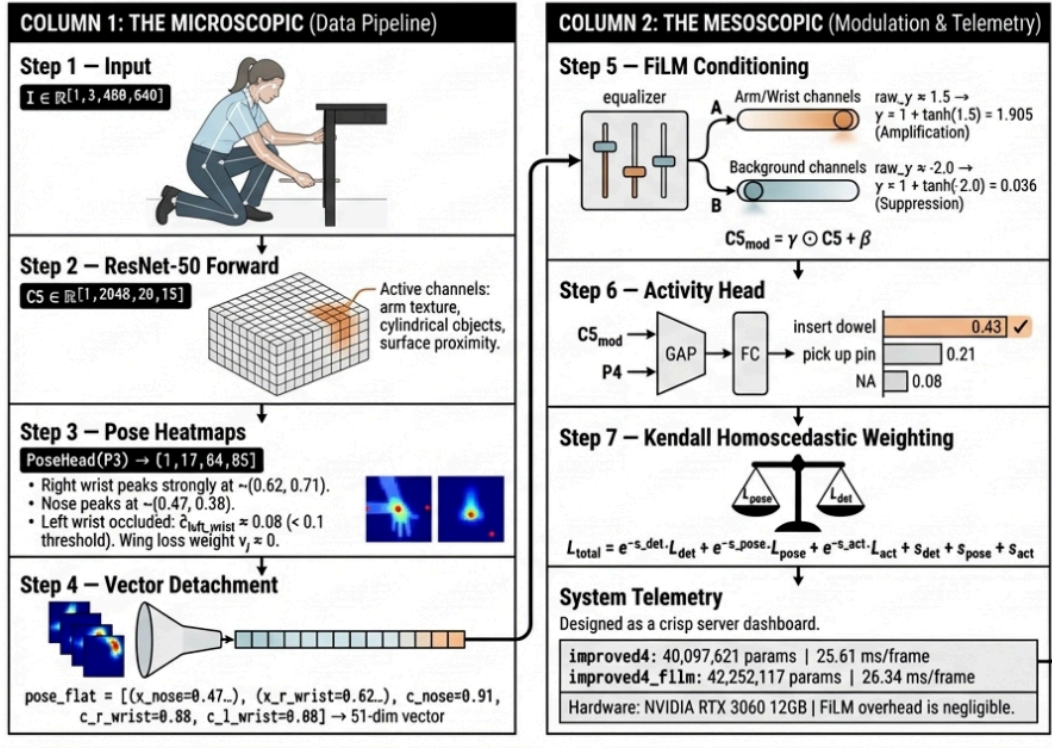
$$L = -\frac{1}{\sigma^2} \log p_y + \log \sigma$$

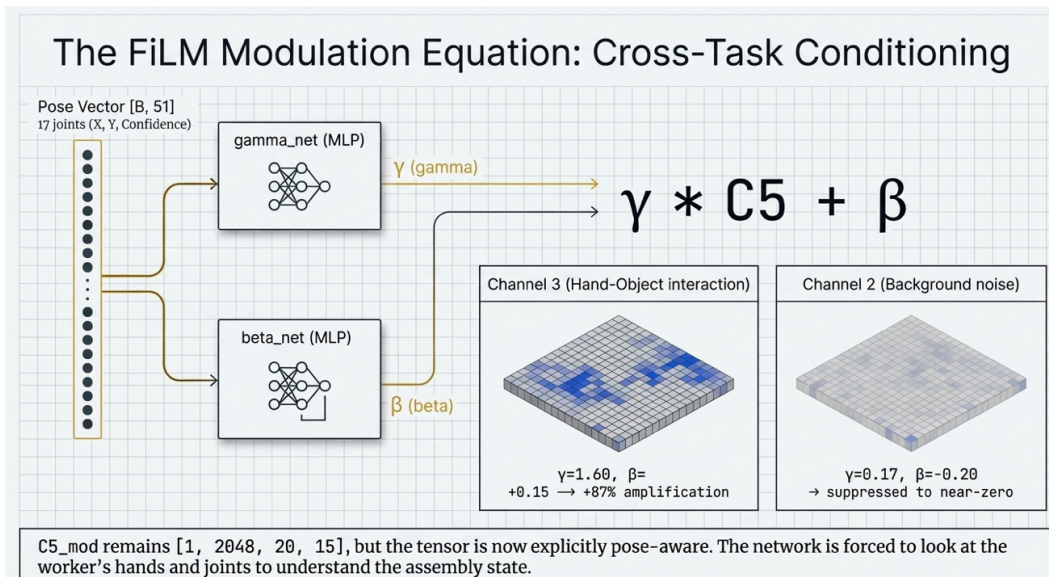
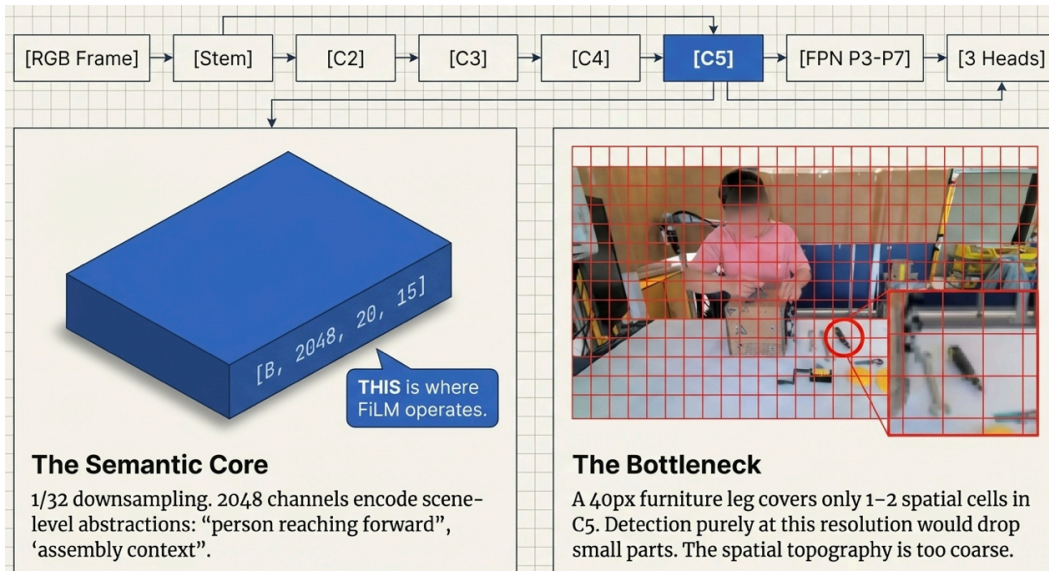




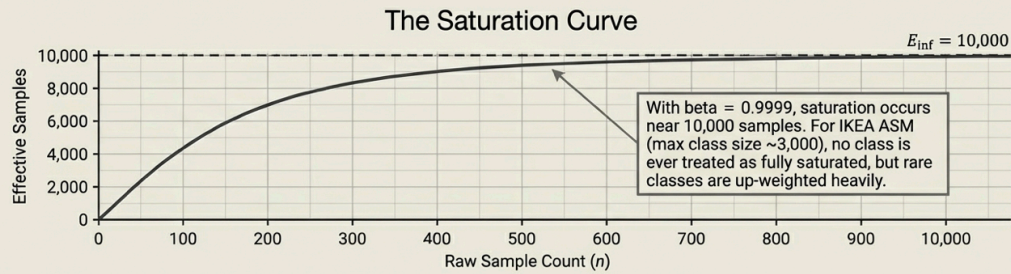
Checkpoint	Act Top-1 %	PCK@0.05 %	PCK@0.01 %	Det mAP@0.5 %	ms/frame	Params
improved3	20.69	92.48	98.18	59.07	25.65	40.1M
improved3_film	32.29	94.27	98.34	59.75	26.29	42.3M
improved4	37.92	99.53	99.89	53.53	25.61	40.1M
improved4_film	37.42	99.58	99.89	60.02	26.34	42.3M

$$H_{\text{out}} = (H_{\text{in}} - 1) \times s - 2p + k = (80 - 1) \times 2 - 2 + 4 = 160$$





The Math of Balance (Class-Balanced Weights)



Diagnostic Table

"Rare Action"	$n_c = 50$	$\alpha = \frac{0.0001}{1 - 0.9999^{50}}$	$\alpha = \frac{0.0001}{0.00499}$	$= 0.02005$
"Medium Action"	$n_c = 176$	$\alpha = \frac{0.0001}{1 - 0.9999^{176}}$	$\alpha = \frac{0.0001}{0.01726}$	$= 0.00579$
"Spin Leg"	$n_c = 683$	$\alpha = \frac{0.0001}{1 - 0.9999^{683}}$	$\alpha = \frac{0.0001}{0.06660}$	$= 0.00150$

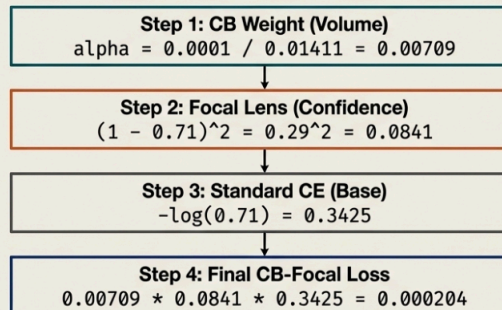
The Result:

$\frac{\alpha(50)}{\alpha(683)} = \frac{0.02005}{0.00150} = 13.4\times$
 The rare class automatically receives 13.4x more gradient weight per sample, mathematically leveling the precision scale.

The Unified Cure: A Worked Example

$$L = \alpha * (1 - p_y)^{\gamma} * (-\log p_y)$$

Target Frame Data | True Class: insert_dowel | Class Size (n_c): 142 | Predicted Probability (p_y): 0.71



Synthesis Insight: For a moderately-confident prediction on a rare class, the focal factor reduces the loss by 12x compared to standard CB alone. This dual-action mechanism mathematically frees gradient capacity, forcing the network to learn uncertain assembly steps.

Real-World Viability

Baseline Speed: 25.61 ms/frame (39 FPS)

FiLM Speed: 26.34 ms/frame (38 FPS)

Hardware Spec: NVIDIA RTX 3060 12GB

POPW (improved4_film): Pose-Conditioned Multi-Task Architecture for IKEA Assembly Recognition

